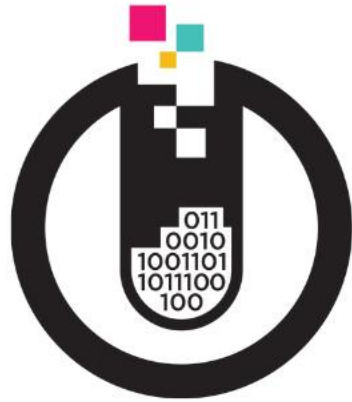


DATA:Scotland 2025



**THE
DATA LAB**
value from data



DATAmasterminds



**ADVANCING
ANALYTICS**



Microsoft Fabric: SQL Databases

Kamil Nowinski





Kamil Nowinski

Microsoft Data Platform **MVP**, **Databricks Champion**
Speaker, blogger, **YouTube**r, data enthusiast

Group Manager at [Avanade](#) UK&I
>20 years of experience in IT

Founder of blog [AzurePlayer](#)

GitHub: **#adftools**, SCD Merge Wizard and more...

Member of the Data Community PL

Former co-organiser of SQLDay (PL), Data Relay (UK),
volunteer at SQLBits (UK)

SQL Server Certificates:

MCITP, MCP, MCTS, MCSA, MCSE Data Platform,

MCSE Data Management & Analytics, DevOps Expert

Moreover: Bicycle, Running, Digital photography, Aikido

@AzurePlayer.bsky.social

Blog



Azure Player

Play with data & have fun!

- Technical posts
- Various skill level
- Cheet sheets
- Recommended books
- Many useful other links
- Interviews (Podcast)
- YouTube Channel:

www.AzurePlayer.net/YouTube



Scan me

www.AzurePlayer.net



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This set of videos promote episodes of "Ask SQL Family" podcast.



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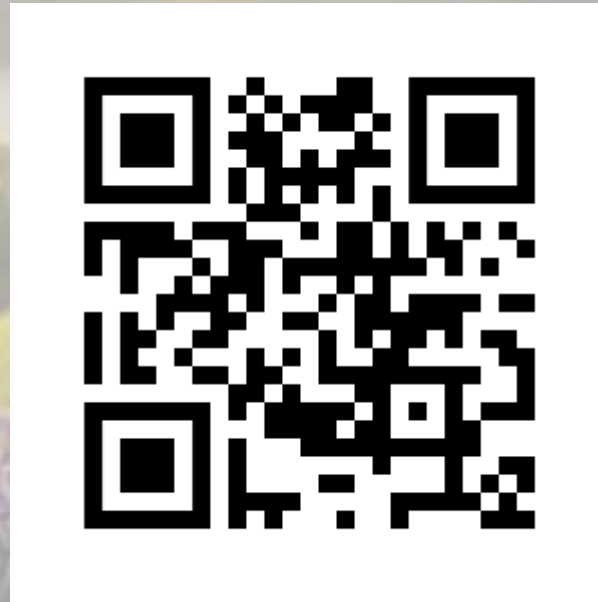


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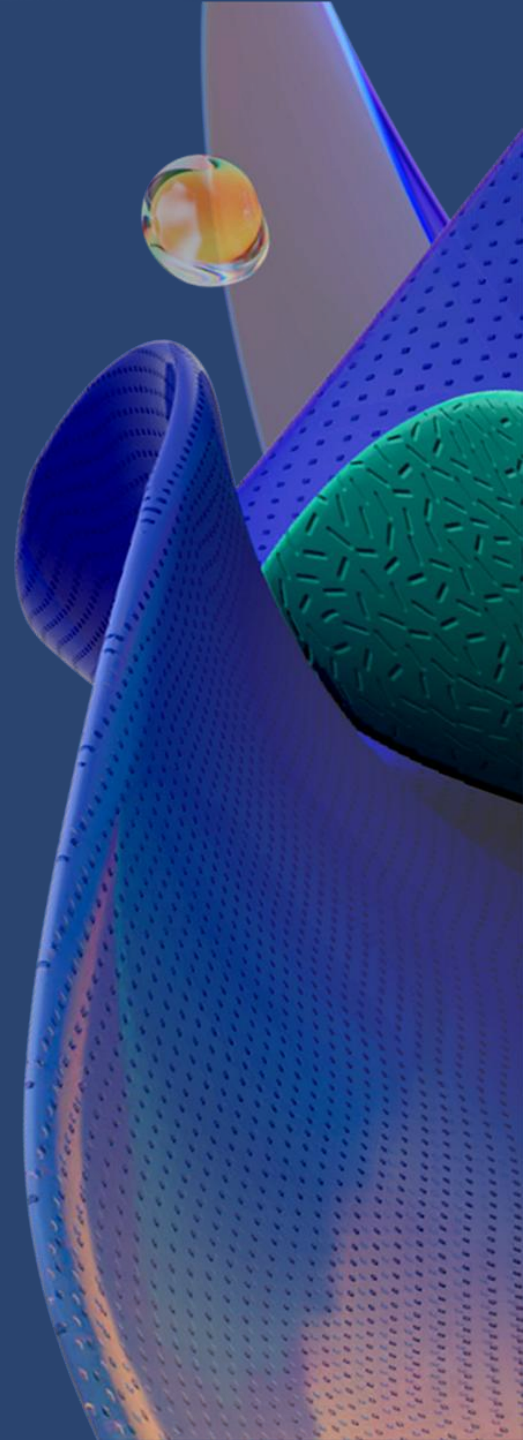
Slides available:



<https://azureplayer.net/slides>

Agenda

- What's Microsoft Fabric – very quick overview
- New Fabric offering: SQL Database
- Architecture
- Demo
- Autonomous & Secure
- Summary



Database users want
the same unified,
simple SaaS experience
offered by Fabric



Quickly spin-up databases



Automate database
management



Seamlessly integrate with
analytics and AI



Microsoft Fabric

The unified data platform for AI transformation



Data
Factory



Analytics



Databases



Real-Time
Intelligence



Power BI



AI



OneLake



Purview



Microsoft Fabric

The unified data platform for AI transformation

Public Preview



Data
Factory



Analytics



Databases



Real-Time
Intelligence



Power BI



AI



OneLake



Purview

Microsoft SQL in the cloud

A unified SQL portfolio built on the industry-leading SQL Server engine

SQL Server on Azure
Virtual Machines



Best for re-hosting and apps requiring OS-level access and control
Automated manageability features and OS-level access

Infrastructure as a Service

Azure SQL Managed
Instance



Best for modernizing existing apps
Offers high compatibility with SQL Server and native VNET support

Platform as a Service

Azure SQL
Database



Best for building new apps in Azure
Pre-provisioned or serverless compute and Hyperscale storage to meet demanding workload requirements

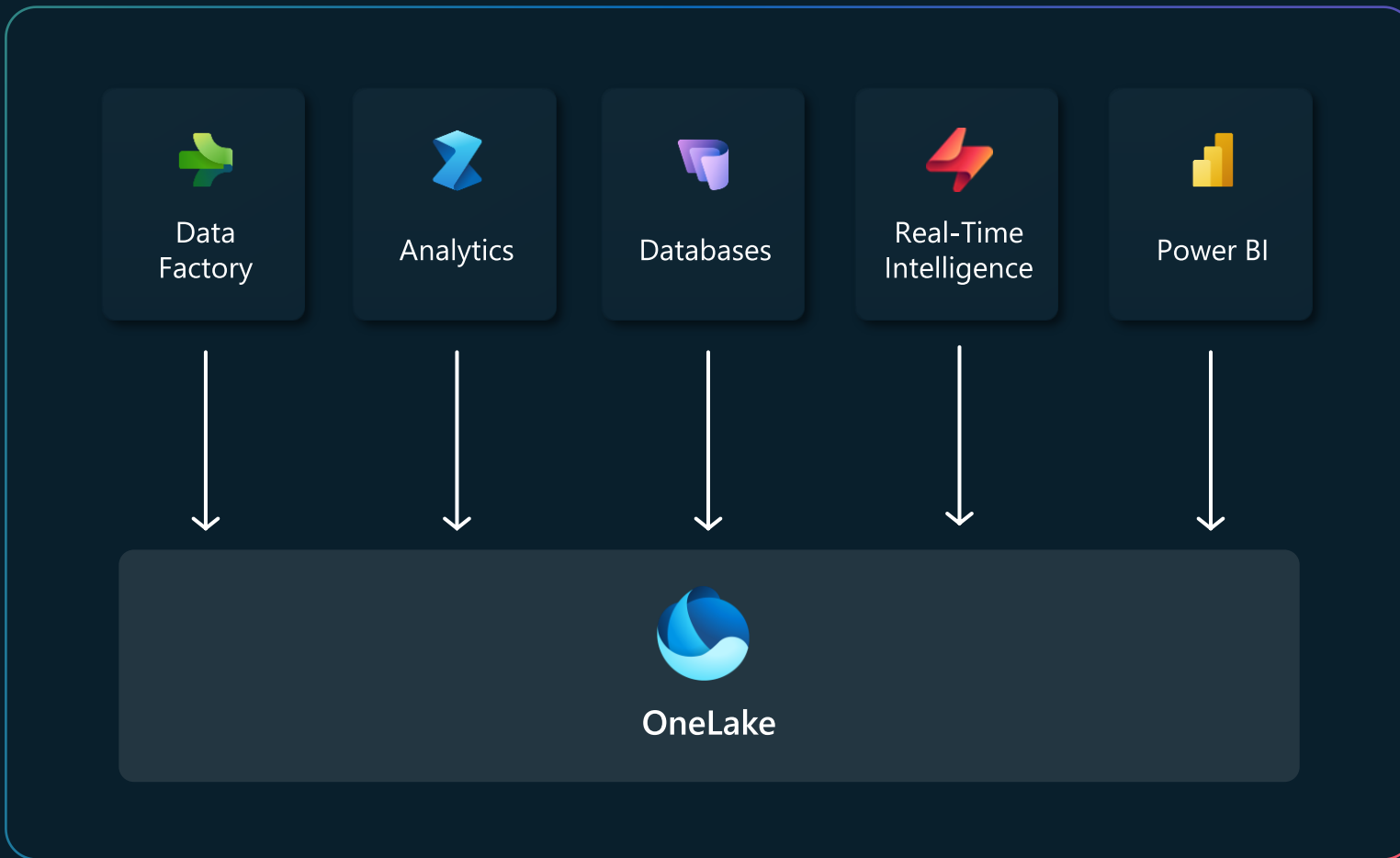
SQL Database in
Fabric



Best for SaaS experience
OLTP to insights in real-time
AI-assisted, simplified, autonomous platform
Deeply integrated with Fabric
Database source control

Software as a Service

OneLake is the OneDrive for data



Single SaaS lake for the whole organization

Single open format used by all workloads to store and query data

Automatically indexed for discovery, MIP labels, lineage, PII scans, sharing, governance and compliance

Focus on building your app

A simple, integrated developer experience



Easy provisioning



AI assistance



Integrations from the editor



T-SQL support

The screenshot displays the Microsoft SQL Server Data Studio interface. The top navigation bar includes tabs for Home, Replication, and Security. Below this is a toolbar with icons for Get data, New Query, Templates, Open in, New API for GraphQL, Performance summary, and Copilot. The Explorer pane on the left shows a tree view of the database structure: ContosoDB > dbo > Tables. The main area displays a 'Data preview - Products' table with 18 rows. The table columns are ProductID, ProductName, Description, Price, Category, and StockQuantity. The status bar at the bottom indicates 'Succeeded (3 sec 17 ms)'.

	ProductID	ProductName	Description	Price	Category	StockQuantity
1	1	ScreenGuard Pro	Tempered glass s...	7.28	Accessory	79
2	2	CaseMate X	Durable phone ca...	6.19	Accessory	8
3	3	PowerBank Z	Portable power b...	11.13	Accessory	28
4	4	CarMount T	Car mount for ph...	19.76	Accessory	80
5	5	WirelessCharger S	Wireless charging...	12.73	Accessory	76
6	6	BluetoothSpeaker R	Portable Bluetoot...	44.51	Accessory	31
7	7	SelfieStick V	Extendable selfie ...	6.44	Accessory	25
8	8	PhoneStand A	Adjustable phone...	8.69	Accessory	32
9	9	MemoryCard M	High-capacity me...	45.53	Accessory	81
10	10	Earbuds E	Wired in-ear earb...	6.36	Accessory	100
11	11	Stylus Pen	Precision stylus pen	7.55	Accessory	52
12	12	CleaningKit C	Phone cleaning kit	9.67	Accessory	15
13	13	LensKit L	Clip-on camera le...	31.75	Accessory	32
14	14	VRHeadset V	Virtual reality hea...	50.49	Accessory	26
15	15	GameController G	Bluetooth game c...	46.47	Accessory	24
16	16	SmartWatch S	Smartwatch with f...	115.02	Accessory	96
17	17	FitnessBand F	Fitness tracking b...	69.02	Accessory	53
18	18	USBHub U	Multi-port USB hub	23.89	Accessory	11

Replication to OneLake

Near real-time replication to OneLake by default, seamless integrations with the rest of Fabric



Unified operational and analytics



Data replicated to open-source Delta



Shortcuts and SQL analytics endpoint to query data without copying it

	Name	Type
	ContosoDB 	SQL database
	 ContosoDB	Semantic model (default)
	 ContosoDB 	SQL analytics endpoint

SQL analytics endpoint ▾



SQL Database
Define, query, and update data



SQL analytics endpoint
Query data using SQL

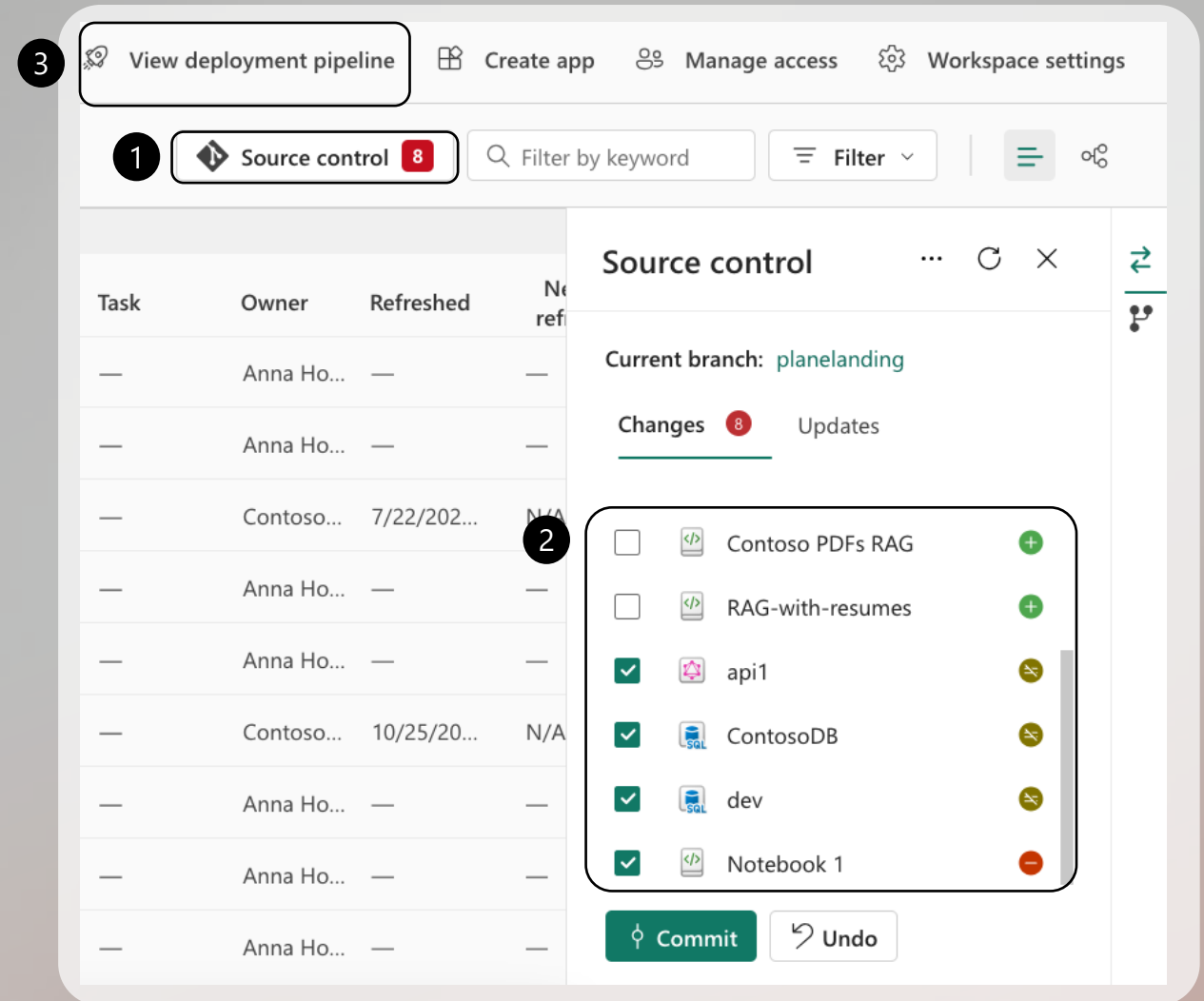
Application lifecycle management

Every database under source control

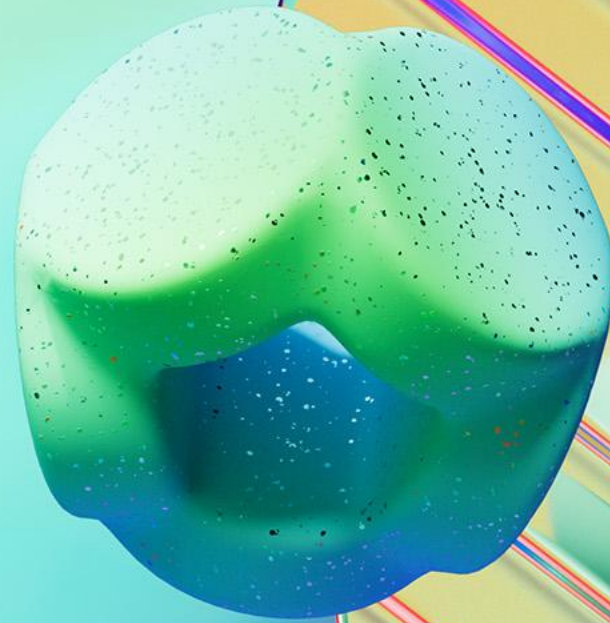
Source control integration in the Fabric web interface (GitHub and Azure DevOps)

Database code editing in developer tooling

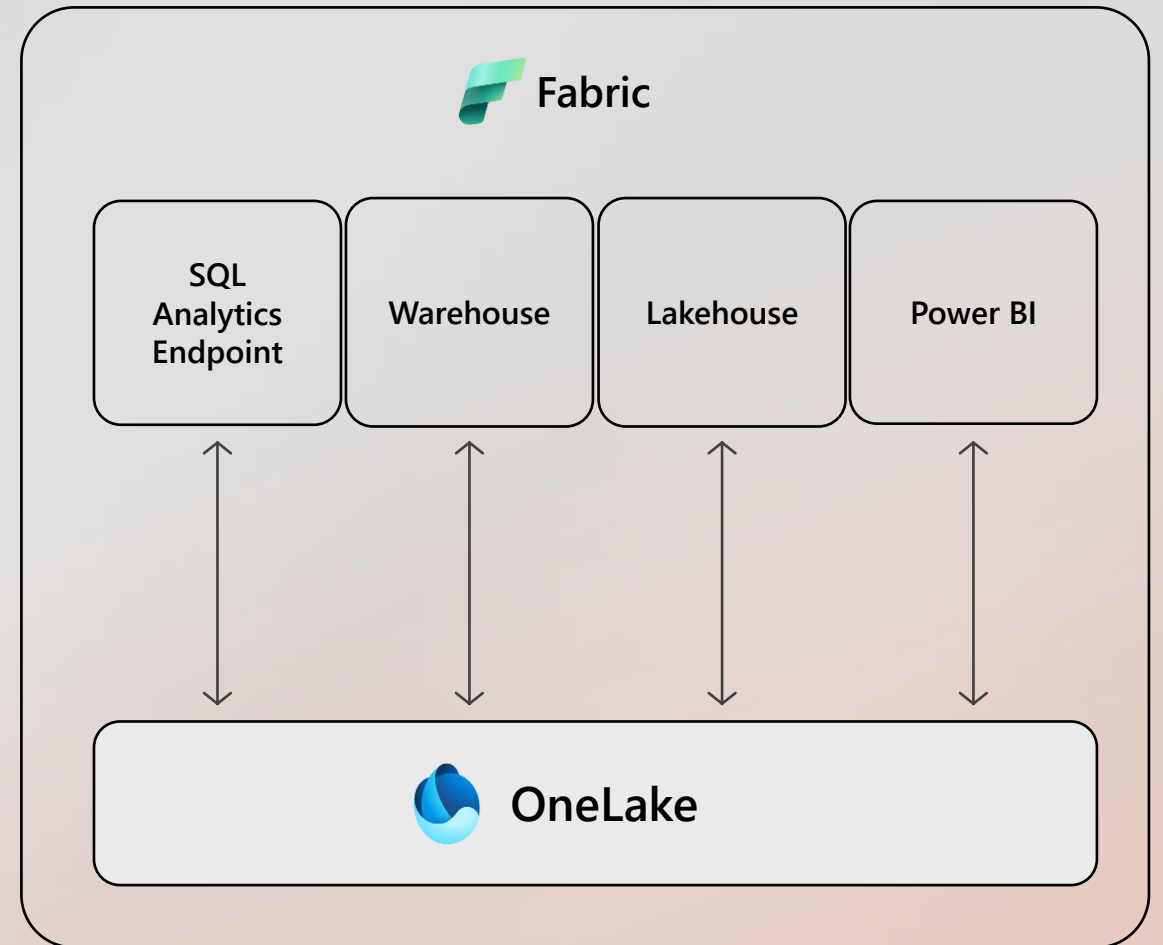
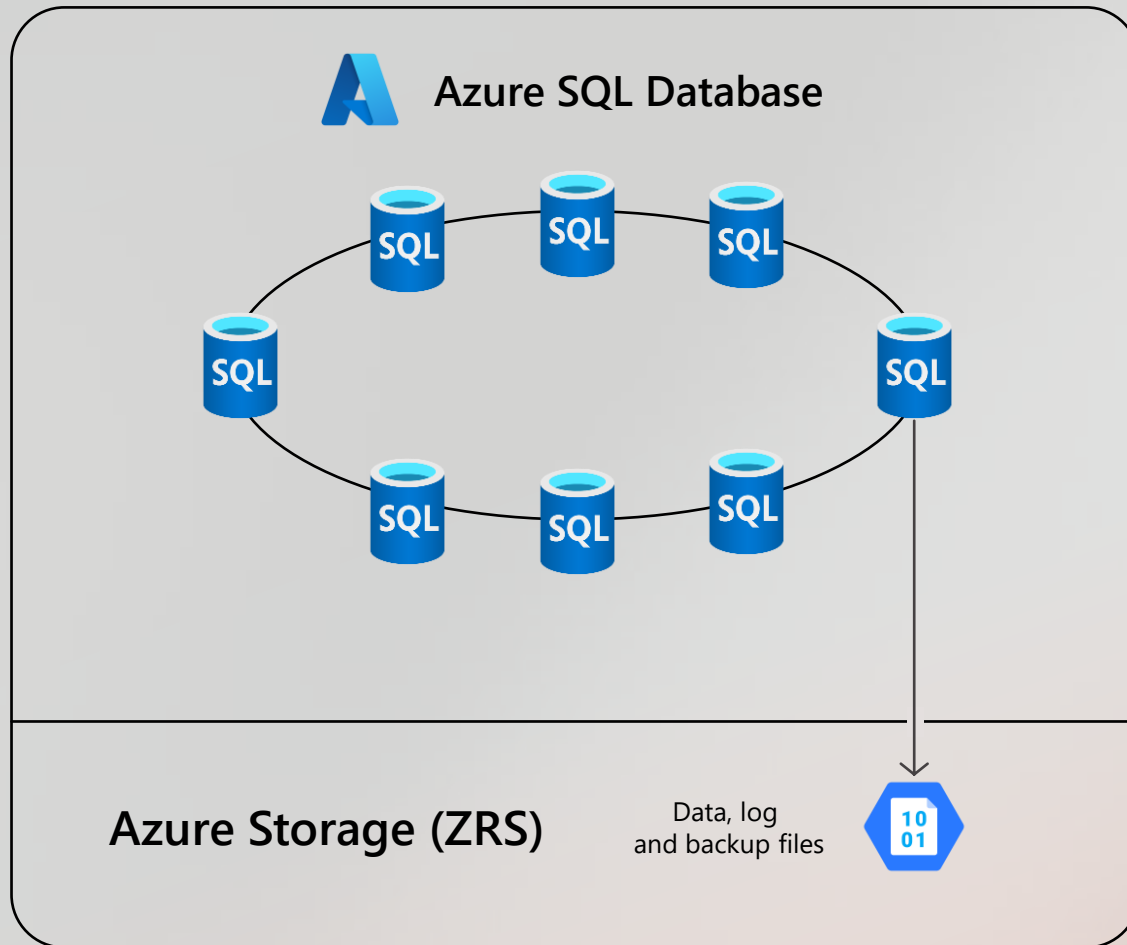
SqlPackage command line tool



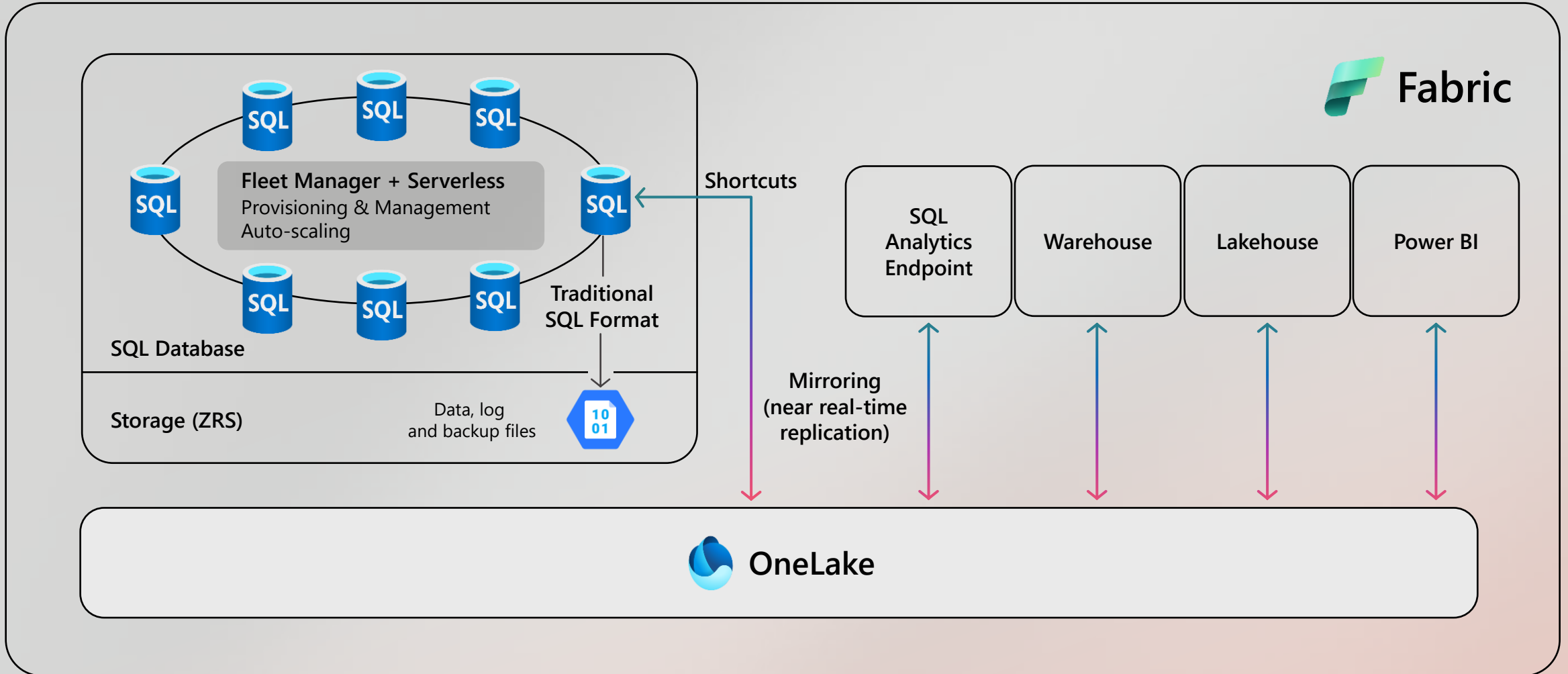
Architecture



Architecture



Architecture



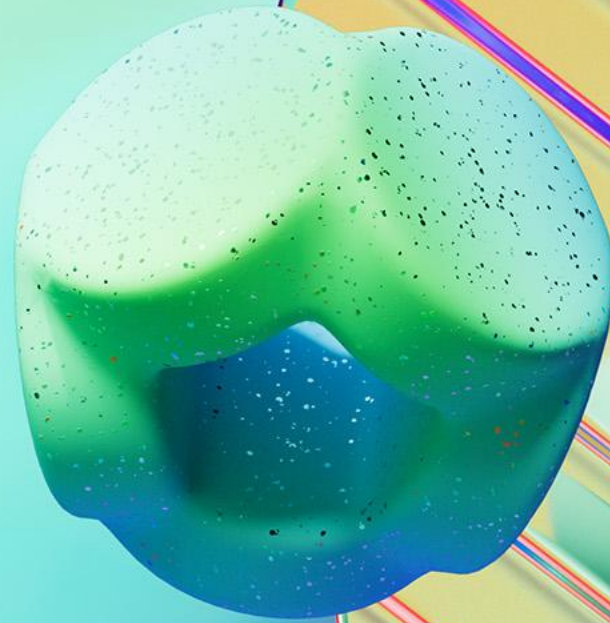
The power of Fabric with no compromises on your SQL operational database

Demo



SQL Database

**Autonomous
and secure**



Autonomous capabilities manage your workloads



Challenge

Making the right choices and managing databases is complex.



Solution

SQL database in Fabric enables you to get the best security, availability, and performance configurations

Secure by default

Auto-managed HADR

Auto-scaling and auto-optimized



Secure and governed by default

Authentication

Microsoft Entra authentication **only**

- Central identity management
- Password-less authentication

Authorization

Workspace roles and item permissions for easy management

Proven SQL native features for granular access control

Network security

Private link-ready

All data encrypted in transit
- TLS 1.2

Data encryption

All data encrypted at rest

Customer Managed Key support

Auditing*

Enabled by default and customizable

Central governance

Microsoft Purview sensitivity labels

Centrally-managed protection policies

* SQL analytics endpoint and Warehouse in Microsoft Fabric

Autonomous



Auto-HA with Zone Redundancy

Auto-scaling compute and storage

Auto-replicated to OneLake (no ETL)

Auto-indexing

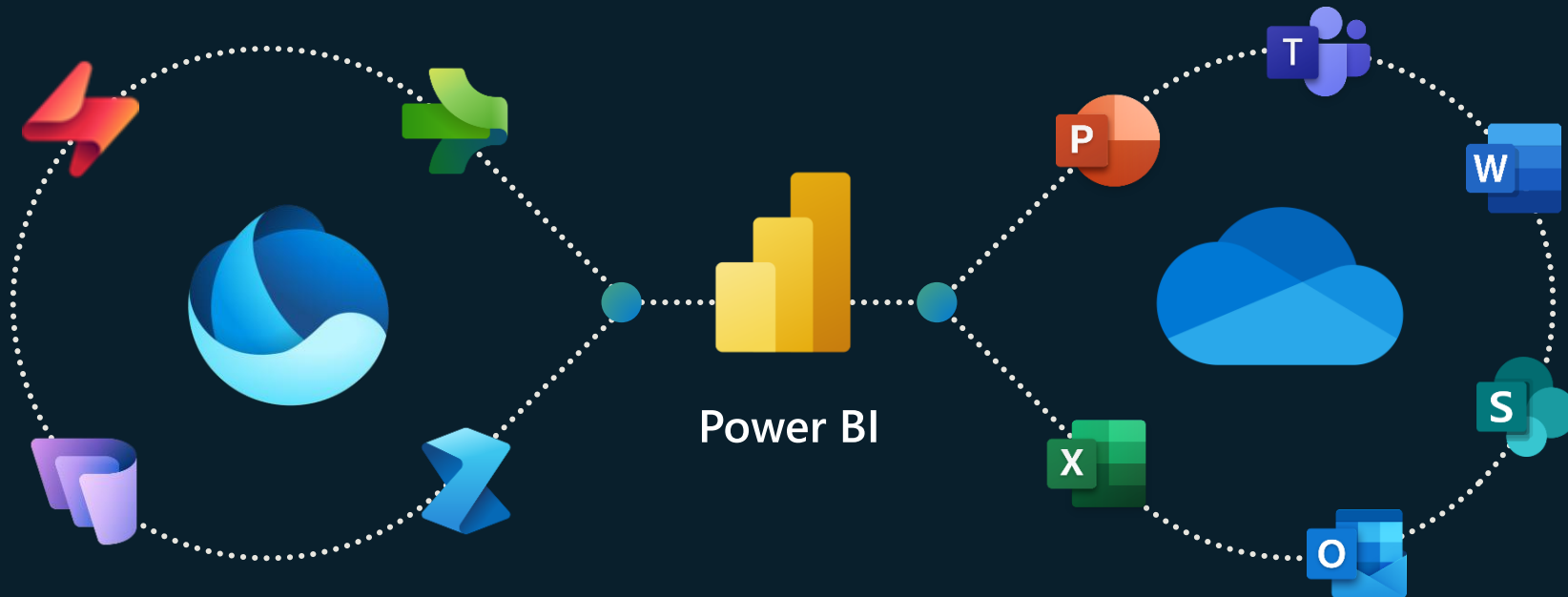
Latest Intelligent Query Processing enhancements

Auto-backups, point-in-time restore available

Auto-patching

Summary

Unified data culture



Unified data culture

Transactional

Analytical

Unified data culture

Translytical

Resources

Get started and engage with the community
to accelerate your AI app development

Explore the documentation and build with our end-to-end tutorials:

<https://aka.ms/fabricsqldocs>

Skill up with our new learn modules: <https://aka.ms/learnfabricsql>

Watch episodes on Data Exposed: <https://aka.ms/dataexposedyt>

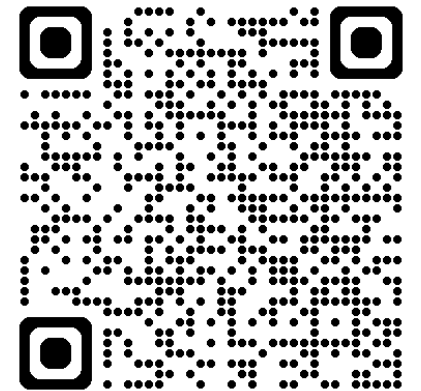
Ask questions, vote for ideas, and read the latest product news:

<https://community.fabric.microsoft.com/>

Live learning series starts December 3rd: <https://aka.ms/LearnFabricDB>

Join us at FabCon Las Vegas in March 2025: <https://aka.ms/FabCon25>

Get started today
with the 60-day
Fabric free trial



Resources

- Explore [Documentation](#) & [End-to-end tutorials](#)
 - [Limitations](#)
 - [Data Exposed episodes](#)
 - [Microsoft Mechanics episode](#)
 - [Join the community](#) for a free live learning series starting
-
- Get the latest product news here on the [Fabric updates blog](#).
 - [My Fabric Labs series](#) (YouTube playlist)
 - [Microsoft Fabric Community](#) (My YouTube playlist)
 - [My GitHub-Fabric list](#)
 - [FabricTools](#) – PowerShell module

Learn with AzurePlayer



Essentials of database projects with SSDT



Database projects with SQL Server Data Tools (SSDT)

<https://learn.azureplayer.net/>

Session Feedback



Event Feedback

DATA:Scotland

DATA:Scotland

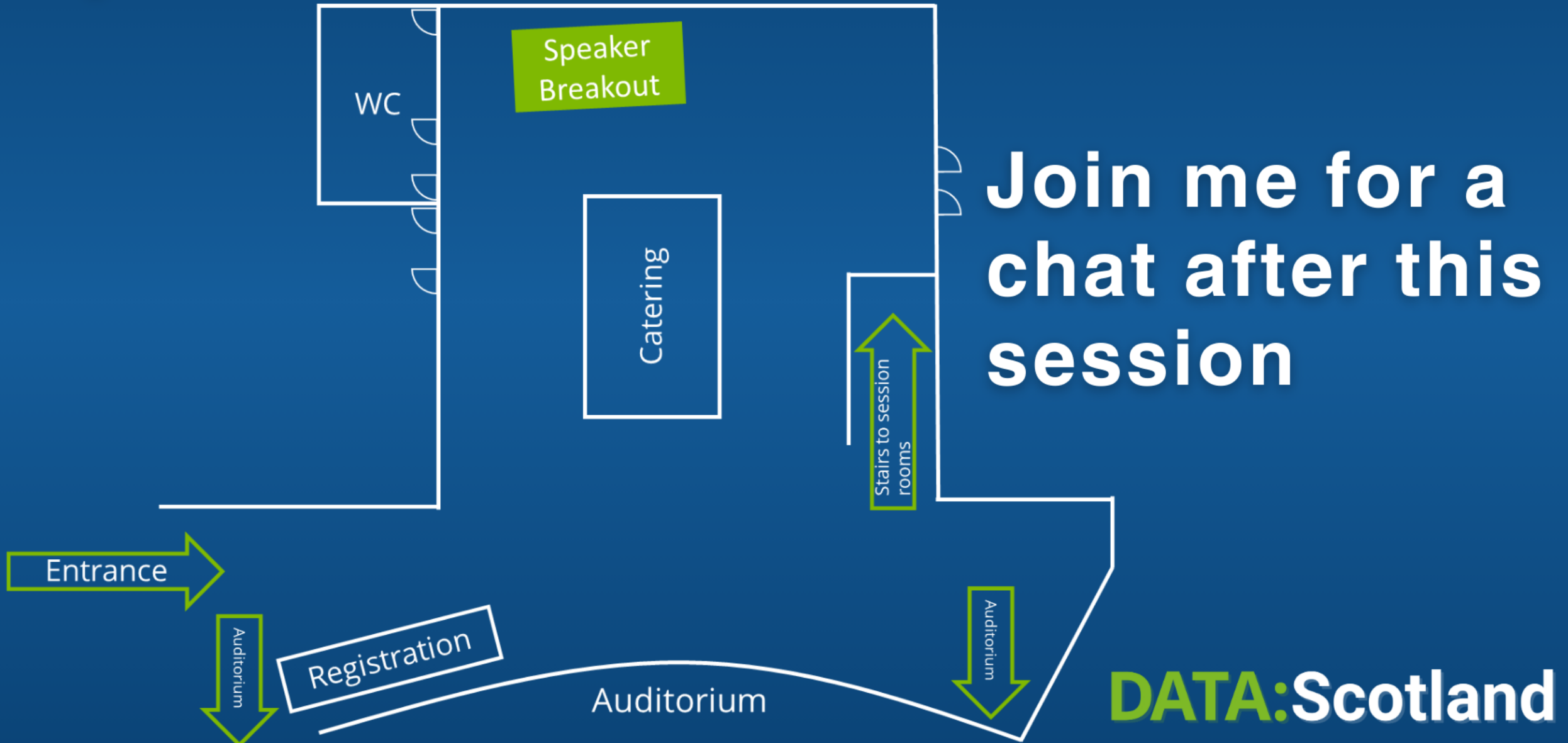


**ADVANCING
ANALYTICS**

After Party

Brewdog, Merchant City | 5pm till late

Speaker Breakout



Thank you!



kamil@azureplayer.net



@AzurePlayer.bsky.social



<https://AzurePlayer.net/slides>



kamilnowinski



AzurePlayer.net

Kamil Nowinski

Microsoft Data Platform MVP, Databricks Champion
Analytics Architect, Azure DevOps Engineer Expert

